ModelIt!

Student Activity Pack

The Secret War in Your Backyard

How Ecosystems Balance Competition, Cooperation, and Survival

Grade 6 | NGSS MS-LS2-2, MS-LS2-3, MS-LS2-5 | Stability and Change

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What living things can you find in a typical backyard? List as many as you can.

2 What do you think would happen if all the birds in an area suddenly disappeared?

3 Why do you think some species go extinct while others thrive?

4 Draw a food chain that might exist in your backyard.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Predator Population | Prey Population | Habitat Size | Resource Availability

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Balanced Ecosystem

Moderate predators, prey, habitat, and resources

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Predator Removal

Predator Population locked at zero

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Habitat Loss

Habitat Size reduced to 25%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Invasive Species — Non-native organisms that outcompete local species for resources, often with no natural predators

Decomposer Activity — The rate at which dead organisms are broken down and nutrients recycled back into the soil

Climate Conditions — Temperature and rainfall patterns that determine which organisms can survive in an area

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Even a tiny backyard is a hidden battlefield — how do organisms compete, cooperate, and survive in a complex ecosystem? Use evidence from your simulation to support your answers.

1. **Why do predator and prey populations rise and fall in cycles instead of staying constant?**

2. **What happened when wolves were removed from Yellowstone — and what happened when they were brought back?**

3. **Why does losing biodiversity make an ecosystem more vulnerable to collapse?**

4. **Is it possible for an ecosystem to survive without decomposers? Why or why not?**

5. **How does your backyard ecosystem connect to ecosystems around the world?**

8 STEM Challenge: Design a Biodiversity Rescue Plan

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YOUR ENGINEERING MISSION

Design an evidence-based plan to restore biodiversity in a degraded ecosystem, using your model to predict which interventions will be most effective.

THE SCENARIO

A local nature preserve has lost 60% of its habitat to development. Deer populations have exploded without wolves, overgrazing vegetation. Bird species are declining, and the stream ecosystem is degrading. Your ecology team must design a restoration plan using model evidence.

Engineering Design Thinking

1. WHICH SPECIES WILL YOU PRIORITIZE FOR PROTECTION OR REINTRODUCTION?
2. HOW MUCH HABITAT NEEDS TO BE RESTORED AND WHAT TYPE?
3. WHAT TIMELINE DO YOU PREDICT FOR ECOSYSTEM RECOVERY BASED ON YOUR MODEL?
4. HOW WILL YOU MEASURE WHETHER YOUR PLAN IS WORKING?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Conservation Biologists and Ecologists study ecosystems to protect biodiversity and restore degraded habitats. They develop strategies that balance human needs with environmental health. They earn $55,000–$110,000/year.